

Analytical MCQ Assessment

AI Generated Paper

SECTION A: Multiple Choice Questions

Q1. Packet filtering firewalls operate primarily at which layer of the OSI model? (1 Marks)

- A. Physical Layer
- B. Data Link Layer
- C. Network Layer
- D. Presentation Layer

Q2. Which type of firewall maintains a state table to track active TCP handshakes and connection states? (1 Marks)

- A. Stateless Packet Filter
- B. Stateful Inspection Firewall
- C. Application-level Gateway
- D. Circuit-level Gateway

Q3. What is a primary disadvantage of an Application-level Gateway (Proxy Firewall) compared to a Packet Filtering Firewall? (1 Marks)

- A. It cannot inspect packet payloads.
- B. It introduces significant performance overhead.
- C. It does not hide internal IP addresses.
- D. It operates only at the Physical layer.

Q4. A circuit-level gateway operates at which layer of the OSI model to monitor TCP handshaking? (1 Marks)

- A. Network Layer
- B. Session Layer
- C. Application Layer
- D. Physical Layer

Q5. Which type of firewall combines traditional firewall capabilities with deep packet inspection (DPI) and intrusion prevention systems (IPS)? (1 Marks)

- A. Next-Generation Firewall (NGFW)
- B. Stateless Firewall
- C. Circuit-level Gateway
- D. Packet Filter

Q6. How does a stateless packet filtering firewall make its forwarding decisions? (1 Marks)

- A. By analyzing the entire application-layer payload.

- B. By tracking the state of established connections.
- C. By inspecting individual packets in isolation based on static rules.
- D. By establishing a proxy connection on behalf of the client.

Q7. Which firewall type acts as an intermediary between a client and a server, completely terminating the incoming connection and establishing a new one? (1 Marks)

- A. Packet Filter
- B. Circuit-level Gateway
- C. Proxy Firewall
- D. Stateful Inspection Firewall

Q8. In the context of firewalls, what does 'Deep Packet Inspection' (DPI) refer to? (1 Marks)

- A. Checking only the source and destination IP addresses.
- B. Analyzing the data field (payload) of a packet beyond just the headers.
- C. Measuring the physical depth of fiber optic cables.
- D. Verifying the MAC addresses of network interface cards.

Q9. Which of the following is a key vulnerability of a simple packet filtering firewall? (1 Marks)

- A. It is highly susceptible to IP spoofing attacks.
- B. It consumes too much CPU processing power.
- C. It completely blocks all UDP traffic.
- D. It requires a dedicated proxy agent for every protocol.

Q10. What is the primary function of a Web Application Firewall (WAF)? (1 Marks)

- A. To filter network-layer traffic based on IP addresses.
- B. To protect web applications by filtering and monitoring HTTP/HTTPS traffic.
- C. To assign dynamic IP addresses to internal clients.
- D. To encrypt all outgoing email communications.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: C

Rationale: Packet filtering firewalls inspect packets at the Network layer to make forwarding decisions based on IP addresses and port numbers.

Q2. Correct Answer: B

Rationale: Stateful inspection firewalls monitor the state of active connections to dynamically allow or block traffic.

Q3. Correct Answer: B

Rationale: Proxy firewalls inspect full application-layer data, which requires more processing power and introduces latency.

Q4. Correct Answer: B

Rationale: Circuit-level gateways operate at the Session layer of the OSI model to verify the validity of TCP handshakes.

Q5. Correct Answer: A

Rationale: Next-Generation Firewalls integrate standard firewall capabilities with advanced features like DPI, IPS, and application awareness.

Q6. Correct Answer: C

Rationale: Stateless firewalls inspect each packet independently using predefined rules without tracking connection state.

Q7. Correct Answer: C

Rationale: A proxy firewall acts as a middleman, terminating the client connection and initiating a separate connection to the destination server.

Q8. Correct Answer: B

Rationale: DPI inspects the actual data payload of packets to detect malware, spam, or policy violations.

Q9. Correct Answer: A

Rationale: Because stateless packet filters do not track connection state, they are vulnerable to IP spoofing attacks where headers are faked.

Q10. Correct Answer: B

Rationale: A WAF specifically targets application-layer threats directed at web applications, such as SQL injection and cross-site scripting.